

CloudFinOps Sentinel

Autonomous cost optimization and auditing for Google Cloud

Gemini 3.5 Flash-Lite · GenAI SDK

Cloud Run · Firestore · Cloud Scheduler · Secret Manager · Cloud Trace

Technical documentation

github.com/Mgodoyd/CloudFinOps-Sentinel-Agent

CloudFinOps Sentinel

An autonomous FinOps agent for Google Cloud. It inspects a Cloud Run estate on a schedule, works out what each resource costs against what it actually uses, fixes the cheap problems by itself, and opens an approval ticket for the risky ones — then remembers what it did so it never does it twice.

Built on **Gemini 3.5 Flash-Lite** through the **GenAI SDK**, running on **Cloud Run** with **Firestore**, **Cloud Scheduler**, **Secret Manager** and **Cloud Trace**.



Command deck

The problem

Cloud waste is not a hard problem to understand. It is a hard problem to *stay on top of*. A service is provisioned at 4 GiB "to be safe", ships, and never touched again. Someone sets `min_instances=1` to dodge cold starts on a service that serves forty requests a day, and it bills around the clock forever. A disk outlives the VM it was attached to. None of this is difficult to spot — it is just nobody's job this week, and the bill arrives monthly.

Dashboards do not fix that, because a dashboard still waits for a person to look at it and then do the work by hand. This agent looks, decides, and acts.

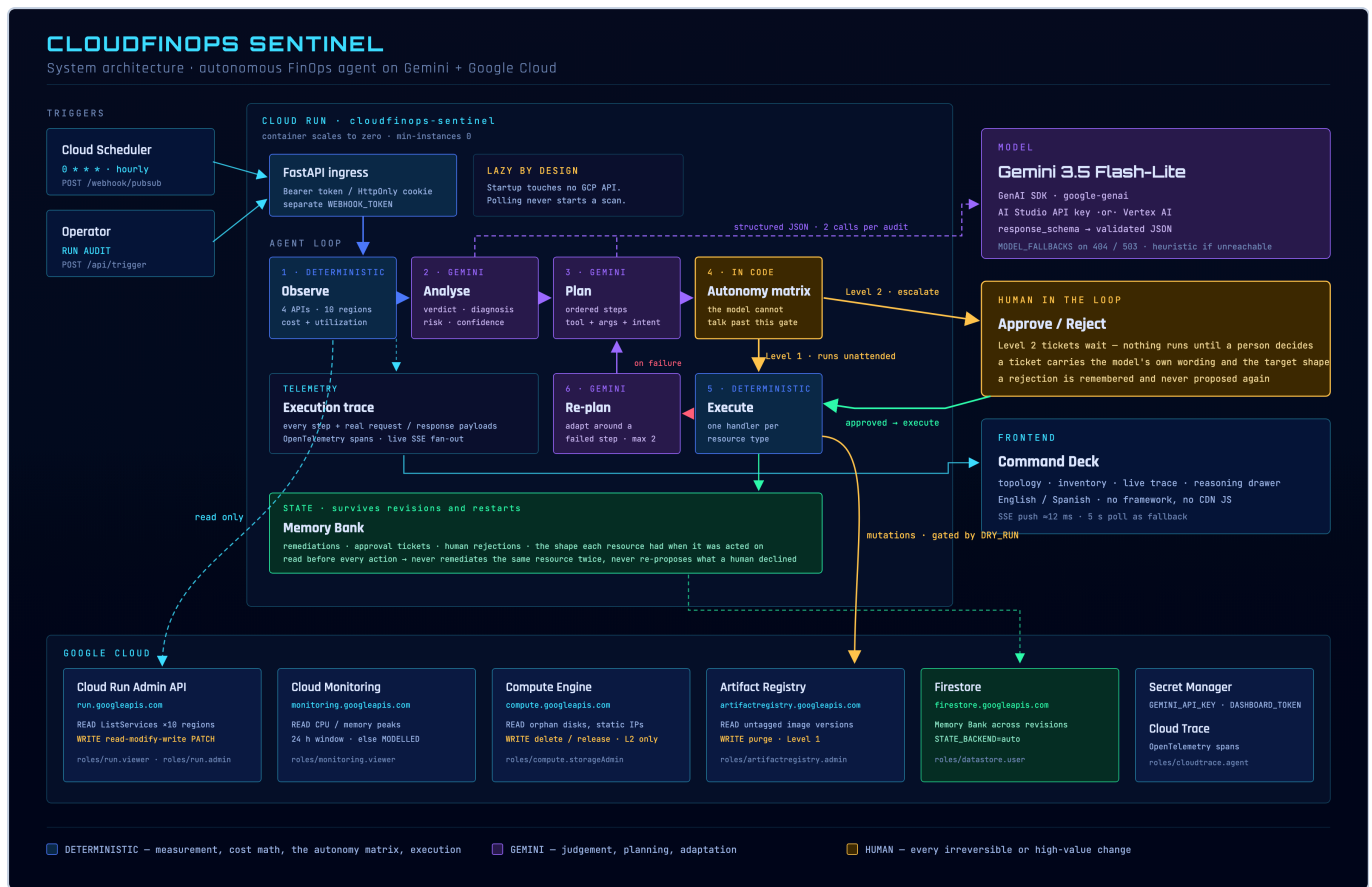
What the agent actually does

Every hour, unattended:

1. **Scans** the project — Cloud Run services across ten regions, orphaned persistent disks, unused static IPs, untagged Artifact Registry images.
2. **Measures** what each one costs and what it actually uses, pulling real CPU and memory peaks from Cloud Monitoring.
3. **Judges** the fleet with Gemini: what is wrong, what shape it should have, what could break if you change it.
4. **Plans** an ordered sequence of tool calls, largest saving first.
5. **Acts** — small, reversible changes are applied directly; high-value or irreversible ones become an approval ticket that waits for a human.
6. **Remembers** every remediation, ticket and rejection, so the next run knows what is genuinely new.

The output is not a report. It is a changed estate, a queue of decisions waiting for a person, and a full audit trail of why.

Architecture



Architecture

The split down the middle is deliberate:

	Produced by	Why
Allocation, utilization, cost, waste	Deterministic code + GCP APIs	Measurement is fact. A model must never invent a number.
Diagnosis, recommendation, risk	Gemini	Judgement is what a model is for.
The plan — which tool, in what order	Gemini	Sequencing actions under a goal.
Adapting after a failed step	Gemini	Deciding whether to retry differently or stop.
Level 1 vs Level 2 vs no action	Deterministic code	A persuasive model must not be able to talk its way into an irreversible action.
Execution	Deterministic code	One handler per resource type, never inferred.

The agent loop

```
Trigger → Observe → Analyse → Plan → Execute → Adapt → Remember
      (code)   (Gemini) (Gemini) (code) (Gemini) (Firestore)
```

Not a chat loop and not a fixed script. Each audit:

1. **Observes** — four GCP APIs across ten regions in parallel, plus Cloud Monitoring for real utilization peaks.
2. **Analyses** — one structured Gemini call over the whole fleet returns a verdict, diagnosis, target shape, risk and confidence per resource.
3. **Plans** — a second Gemini call turns that analysis into an *ordered plan*: which tool, on which resource, with which arguments, and what each step expects to achieve.
4. **Executes** — the agent carries the plan out with the autonomy matrix enforced in code.
5. **Adapts** — when a step fails, the plan goes back to the model to be revised around the failure (up to `MAX_REPLANS = 2`) rather than abandoning the run.
6. **Remembers** — remediations, tickets, runs, and the *shape* each resource had when it was acted on.

Cost is bounded deliberately: **two Gemini calls per audit**, plus one per failed round. A tool-calling loop would spend one request per invocation and exhaust a free-tier minute on a single audit.

Both model calls use `response_schema`, so the SDK validates the shape and there is no parsing guesswork. The planner may only choose from a fixed toolbox (`resize_service`, `delete_disk`, `release_address`, `delete_image`, `skip`) — it cannot invent a tool or an argument name.

The decision model

This is the part worth reading closely: how a measurement becomes an action, and exactly which numbers decide it.

HOW THE AGENT DECIDES

Every threshold and formula that turns a measurement into an action – all enforced in code, not in the prompt

STEP 1 · MEASURED – never produced by the model

Facts about one resource

cpu_limit · memory_limit · min_instances Cloud Run Admin API
cpu_peak % · memory_peak % (24 h window) Cloud Monitoring
when Monitoring is unavailable the peaks are MODELLED and the row is labelled as such

STEP 2 · COST MODEL

What the allocation costs per month

$base = (vCPU \times \$0.000024 + GiB \times \$0.0000025) \times 730 \times 3600 \text{ s}$
 $min_instances > 0$ cost = base × min_instances billed around the clock
 $min_instances = 0$ cost = base × max(cpu_peak, 0.02) billed only while serving
an always-on service with no load is the most expensive kind of idle – that asymmetry is the whole point

STEP 3 · RECOVERABLE WASTE

The share of the bill that buys nothing

$headroom = 1 - \max(cpu_peak, memory_peak)$
 $waste = cost \times \max(0, headroom - 0.20)$ – a 20 % buffer is never counted as waste

STEP 4 · CLASSIFY – is this resource within valid parameters?

Idle min_instances > 0 AND cpu_peak < 10 %
– or – cpu_peak < 10 % AND memory_peak < 20 %

Oversized memory ≥ 1.0 GiB AND headroom > 50 %

Healthy neither rule fires – allocation matches observed usage

Tolerated a rule fired, but waste < \$5.00 – technically idle, practically fine
shown in the estate and the KPIs, never turned into an action

a fleet that always looks broken teaches operators to ignore the colour

a classified resource goes to

STEP 5 · AUTONOMY MATRIX – who is allowed to act

Decided in code. A persuasive model cannot talk its way past it.

waste < \$5.00 NO ACTION · reported only
escalating a \$1 resize costs more attention than it saves

$\$5.00 \leq waste < \40.00 LEVEL 1 · the agent applies it unattended
reversible change, bounded blast radius

waste ≥ \$40.00 LEVEL 2 · approval ticket, nothing runs
a human decides before anything is applied

irreversible tool LEVEL 2 ALWAYS · whatever the saving
delete_disk · release_address · delete_image

STEP 6 · SIZING – the concrete shape proposed

$floor = \max(256 \text{ MiB}, current_memory / 4)$
 $target_memory = \text{smallest valid step} \geq \max(memory_peak \times 1.4, floor)$
 $target_cpu = \text{smallest valid step} \geq \max(cpu_peak \times 1.4, cpu_floor)$

1.4 × observed peak – a 40 % safety headroom over what was actually used
never grows a resource, and never cuts more than 4× in a single audit
a service with minutes of history shows artificially low peaks; a 16× cut on that evidence is a guess, so repeated audits converge instead. When the cap binds, the drawer says so.
valid memory steps 128Mi · 256Mi · 512Mi · 1Gi · 2Gi · 4Gi · 8Gi · 16Gi · 32Gi
valid cpu steps 0.25 · 0.5 · 1 · 2 · 4 · 8 · Cloud Run: ≥1 vCPU for ≥4Gi, ≥2 vCPU for ≥8Gi

EVERY NUMBER ABOVE IS CONFIGURABLE

ENVIRONMENT VARIABLE	DEFAULT
MIN_SAVINGS_THRESHOLD	\$5.00
HIGH_RISK_ROI_THRESHOLD	\$40.00
MEMORY_ANOMALY_THRESHOLD_GIB	1.0
METRICS_LOOKBACK_HOURS	24
DRY_RUN	true – writes disabled

WORKED EXAMPLE – every number below is reproduced by the running agent

checkout-api Cloud Run · us-central1

MEASURED 2 vCPU · 4Gi · min_instances 2 · peaks 20.4 % CPU / 31.4 % memory

cost	$(2 \times 0.000024 + 4 \times 0.0000025) \times 2,628,000 = \$152.42 \text{ per instance} \times 2$	= \$304.84 / mo
headroom	$1 - \max(0.204, 0.314)$	= 68.6 %
waste	$304.84 \times (0.686 - 0.20)$	= \$148.15 / mo
classify	$4Gi \geq 1.0Gi$ AND headroom 68.6 % > 50 %	Oversized
autonomy	$\$148.15 \geq \40.00 threshold	Level 2 · approval ticket
sizing	memory $\max(4096 \times 0.314 \times 1.4 = 1801, \max(256, 4096/4 = 1024))$ → step 2048 cpu $\max(2 \times 0.204 \times 1.4 = 0.571, \max(2/4 = 0.5, 0.25))$ → step 1.0 min_instances is only driven to 0 when the verdict is Idle, so it stays at 2 here	= 2Gi = 1 vCPU

Decision model

Step 1 – What is measured

Value	Source
cpu_limit, memory_limit, min_instances	Cloud Run Admin API
cpu_peak %, memory_peak % over 24 h	Cloud Monitoring

When Cloud Monitoring is unavailable the peaks fall back to a deterministic model and every row is labelled **MODELLED** instead of **MONITORING**. The confidence rating drops to **low** and the drawer says why.

Step 2 – What it costs

Cloud Run Tier-1 on-demand rates, against the *allocated* shape:

$base = (vCPU \times \$0.000024 + GiB \times \$0.0000025) \times 730 \times 3600 \text{ s}$

The billing model then matters more than the shape:

Condition	Monthly cost	Why
<code>min_instances > 0</code>	<code>base × min_instances</code>	Warm instances bill around the clock whether or not traffic arrives
<code>min_instances = 0</code>	<code>base × max(cpu_peak, 0.02)</code>	A scale-to-zero service only bills while it is serving

That asymmetry is the point: **an always-on service with no load is the most expensive kind of idle**, and it is the single largest saving the agent usually finds.

Step 3 — How much of that is recoverable

```
headroom = 1 - max(cpu_peak, memory_peak)
waste    = cost × max(0, headroom - 0.20)
```

The 20 % buffer is never counted as recoverable. A service running at 85 % peak has 15 % headroom and zero reported waste — that is a correctly sized service, not a saving opportunity.

Step 4 — Is this resource within valid parameters?

State	Condition	Colour
Idle	<code>min_instances > 0 AND cpu_peak < 10 %</code> — or — <code>cpu_peak < 10 % AND memory_peak < 20 %</code>	red
Oversized	<code>memory ≥ 1.0 GiB AND headroom > 50 %</code>	amber
Healthy	neither rule fires	green
Tolerated	a rule fired, but <code>waste < \$5.00</code>	green

Tolerated exists because a resource with \$1/month of recoverable waste is correctly sized for practical purposes. Painting it red implies an action that will never be taken, and a fleet that always looks broken teaches operators to ignore the colour. Such resources stay in the topology and the KPIs — they exist and they are fine — with the drawer explaining why no change is proposed.

Rules for the other resource types are simpler, because the waste is total: **ORPHANED_DISK** (unattached), **UNUSED_STATIC_IP** (reserved, unattached), **UNTAGGED_IMAGE** (no tag, cannot be deployed by name).

Step 5 — Who is allowed to act

Condition	Decision
<code>waste < \$5.00</code>	No action. Reported only.
<code>\$5.00 ≤ waste < \$40.00</code>	Level 1 — the agent applies it unattended
<code>waste ≥ \$40.00</code>	Level 2 — approval ticket, nothing runs
Tool is irreversible	Level 2 always , whatever the saving

Irreversible means `delete_disk`, `release_address` and `delete_image`. The order of these checks matters and is deliberate: the *value* threshold is tested first, because escalating a \$0.50 cleanup costs more human attention than it saves, irreversible or not. Only above the threshold does the question "may this run unattended?" arise.

This is enforced in `app/core/executor.py` and `app/tools/gcp_remediator.py` — in code, not in the prompt. If the model asks for a Level 2 action directly, the tool downgrades it to a ticket.

Step 6 — The shape it proposes

```
floor          = max(256 MiB, current_memory / 4)
target_memory  = smallest valid step ≥ max(memory_peak × 1.4, floor)
target_cpu     = smallest valid step ≥ max(cpu_peak × 1.4, cpu_floor)
```

- **1.4x** the observed peak — a 40 % safety headroom over what was actually used.
- **Never grows** a resource.
- **Never cuts more than 4x in one audit.** A service with minutes of traffic history shows artificially low peaks, and a 16x cut on that evidence is a guess rather than a right-sizing. Repeated audits converge safely instead. When the cap binds, the drawer says so.
- Valid memory steps: `128Mi 256Mi 512Mi 1Gi 2Gi 4Gi 8Gi 16Gi 32Gi`. Valid CPU steps: `0.25 0.5 1 2 4 8`. Cloud Run requires ≥1 vCPU for ≥4Gi and ≥2 vCPU for ≥8Gi, and the sizing respects that.
- `min_instances` is only driven to `0` when the verdict is **Idle**.

A worked example

`checkout-api` — 2 vCPU, 4Gi, `min_instances=2`, peaks 20.4 % CPU / 31.4 % memory:

cost	$(2 \times 0.000024 + 4 \times 0.0000025) \times 2,628,000 = \152.42 per instance $\times 2 = \mathbf{\$304.84/mo}$
headroom	$1 - \max(0.204, 0.314) = \mathbf{68.6\%}$
waste	$304.84 \times (0.686 - 0.20) = \mathbf{\$148.15/mo}$
classify	$4Gi \geq 1.0Gi$ and $68.6\% > 50\%$ → Oversized
autonomy	$\$148.15 \geq \40.00 → Level 2, approval ticket
sizing	memory $\max(4096 \times 0.314 \times 1.4 = 1801, \max(256, 1024))$ → step 2048 = 2Gi cpu $\max(2 \times 0.204 \times 1.4 = 0.571, \max(0.5, 0.25))$ → step 1.0 = 1 vCPU

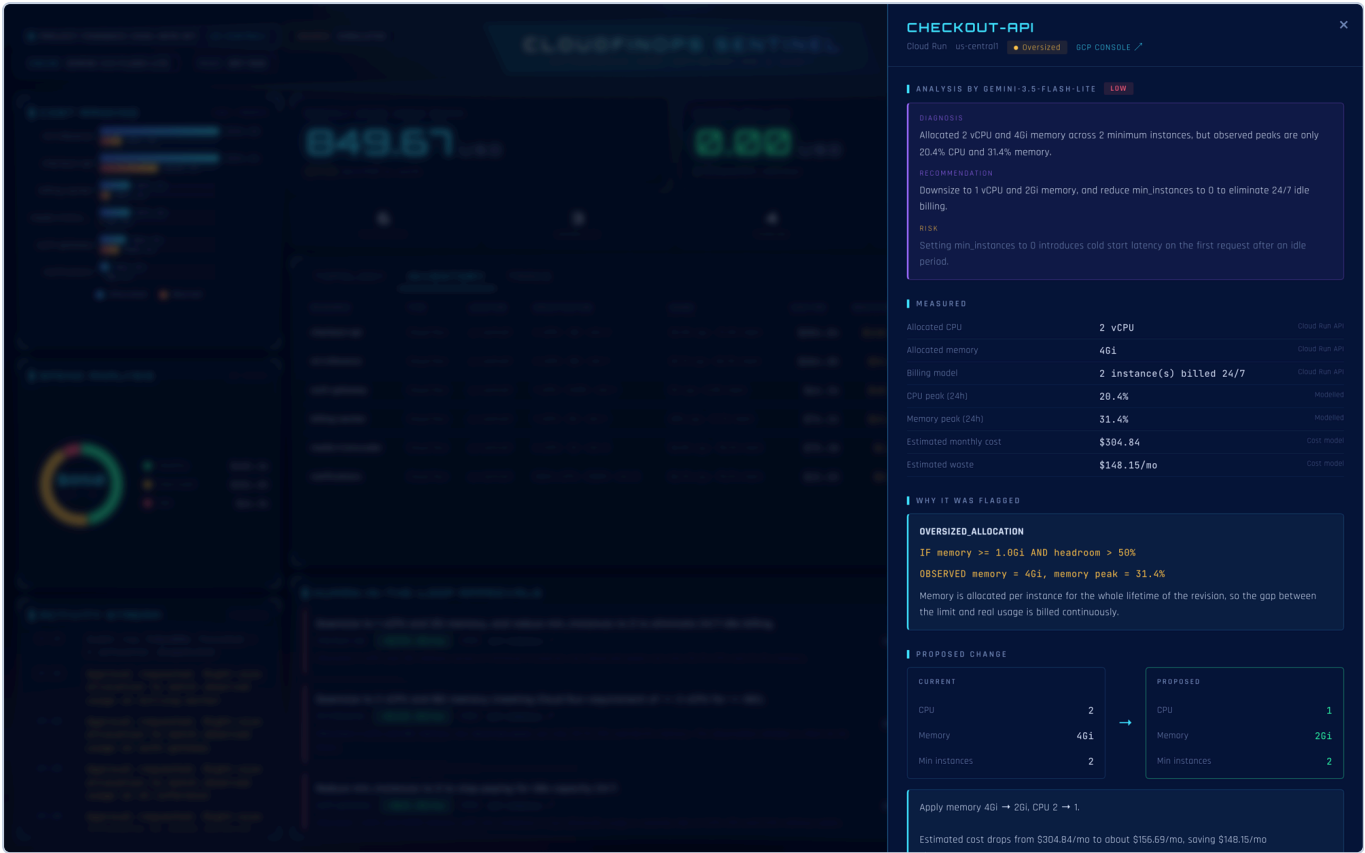
Every threshold is configurable

Variable	Default	Controls
<code>MIN_SAVINGS_THRESHOLD</code>	<code>5.0</code>	Below this, a finding is reported but never actioned
<code>HIGH_RISK_ROI_THRESHOLD</code>	<code>40.0</code>	At or above this, a human must approve
<code>MEMORY_ANOMALY_THRESHOLD_GIB</code>	<code>1.0</code>	Allocation that counts as large enough to be "oversized"
<code>METRICS_LOOKBACK_HOURS</code>	<code>24</code>	Observation window for the peaks
<code>DRY_RUN</code>	<code>true</code>	Master safety gate — writes disabled

Seeing the reasoning

Every recommendation is auditable. Clicking any row, node or approval opens a drawer with the complete chain: the model's judgement first, then the measurements it was given, then the rule, the proposed change, and the autonomy decision.

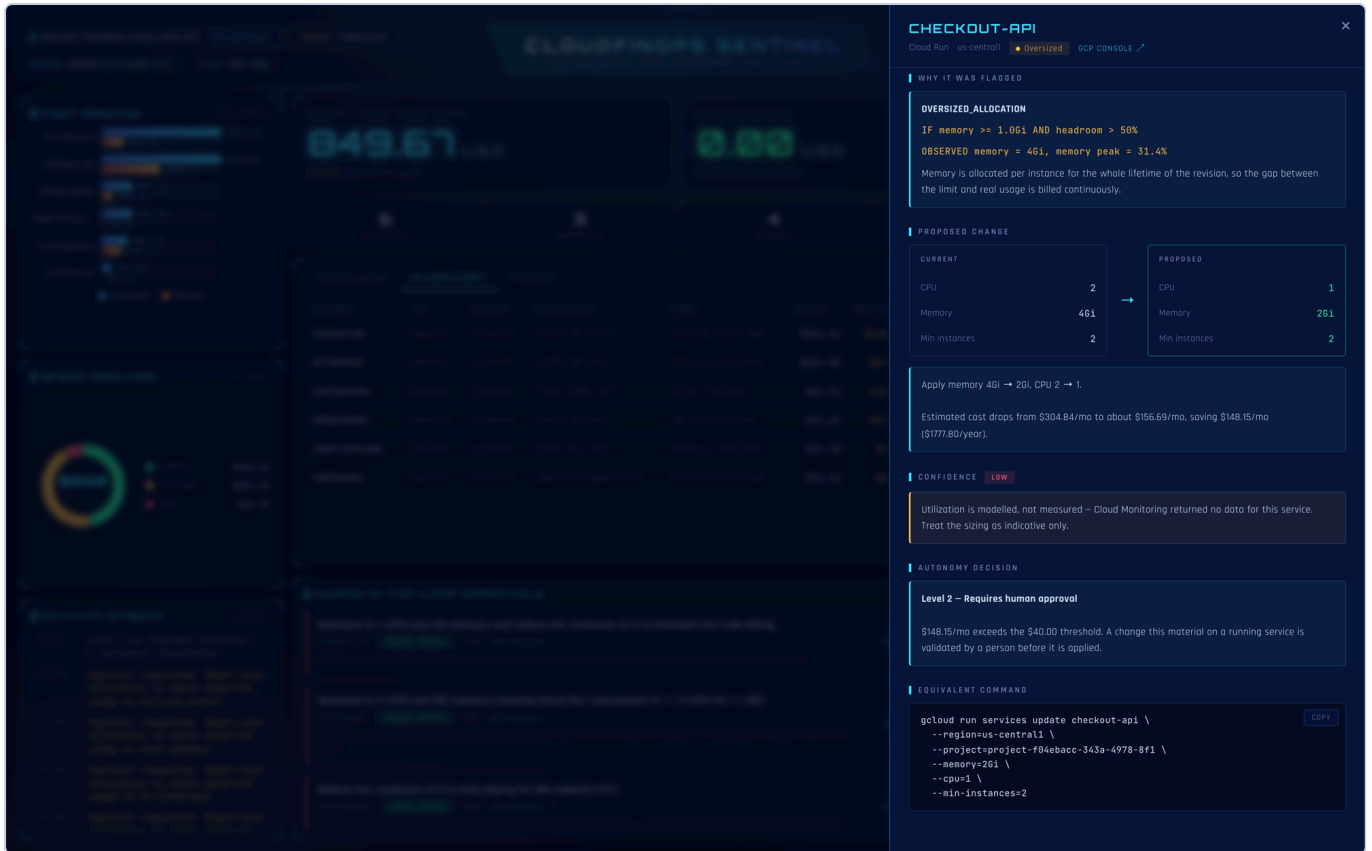
The model's analysis, and the evidence underneath it



Rationale — evidence

The `ANALYSIS BY GEMINI-3.5-FLASH-LITE` block is the model's own wording — diagnosis, recommendation and risk. Below it, `MEASURED` lists every number the model was given, each labelled with where it came from (`Cloud Run API` , `Modelled` , `Cost model`), so a recommendation can always be traced back to the facts behind it. `WHY IT WAS FLAGGED` names the rule id, the condition that was checked, and the values observed.

The proposed change, and why it is allowed or not



Rationale — autonomy

PROPOSED CHANGE is a before/after of the concrete shape, with the monthly and yearly saving. **CONFIDENCE** is graded on how much real metric history exists — **Low** here, because Cloud Monitoring returned no data for this service and the peaks are modelled. **AUTONOMY DECISION** states the level and quotes the threshold that produced it. **EQUIVALENT COMMAND** is the exact `gcloud` an operator could run to apply the same change by hand.

The whole estate, in its own terms



Inventory

The inventory shows every managed resource with its specification, observed usage, monthly cost, recoverable waste and state. Types are described in their own terms — a disk is not forced into Cloud Run's columns.

The autonomy matrix



Approvals and execution

Level 2 findings become tickets in `HUMAN-IN-THE-LOOP APPROVALS`. Nothing runs until someone clicks **APPROVE**.

What you approve is what runs. The headline of every ticket ends in the exact target shape — `→ 1 vCPU / 2Gi / min 0` — and that shape is what execution reads. It is not prose about the change, it is the change:

- The planner names only the dimension it cared about. A step meaning "scale to zero" carries `min_instances` and no memory. What is missing is resolved from the deterministic sizing for that resource, then from the shape it already has — never from a constant, because a default is a change nobody proposed.
- If no shape can be established, the ticket is still raised so the finding stays visible, and execution **refuses** it rather than guessing.
- The shape is validated against Cloud Run's own constraints before it is offered: ≥ 1 vCPU for ≥ 4 Gi and ≥ 2 vCPU for ≥ 8 Gi. A model that says "2 vCPU" in prose and encodes `1` produces a deploy Cloud Run rejects, which reads to an operator as the agent being broken rather than the plan being wrong.
- The saving on the ticket is the **measured** `wasted_cost`, never the model's `estimated_saving`. The model's figure is a guess, and the thresholds and the realised-savings KPI are defined against the measurement.

The shape is deliberately left untranslated, for the same reason a `gcloud` command is: it is a technical value, not a sentence. The model's own wording is kept on the ticket and shown in the reasoning drawer, where it explains the change rather than promising one.

A rejection is remembered. `last_rejection()` is checked before any action, so a change a human has already declined is never proposed again — that repetition is exactly the alert fatigue the savings threshold exists to prevent.

The Memory Bank also records the *shape* a resource had when it was acted on, and a later scan compares against it:

- unchanged since the last action → duplicate, skipped;
- changed and still wasteful → eligible again;
- previously rejected by a human → skipped regardless of shape.

Without the shape comparison a partially fixed resource would be blocked forever, and the dashboard would show an anomaly with no ticket to act on.

Execution and the trace

The **Trace** tab streams every step live over server-sent events, grouped by phase, with the real payloads attached.

The reasoning pipeline, step by step



Trace pipeline

ANALYSIS steps show each resource being evaluated against the detection rules with its recoverable waste. **PLANNING** and **DECISION** show the model building an execution plan and the autonomy matrix classifying each step.

The Gemini call, with its request and response



Trace — LLM call

Clicking any step expands its payload. For a model call that is the request shape, the model used, and how many resources came back analysed.

A mutation, with what was actually sent



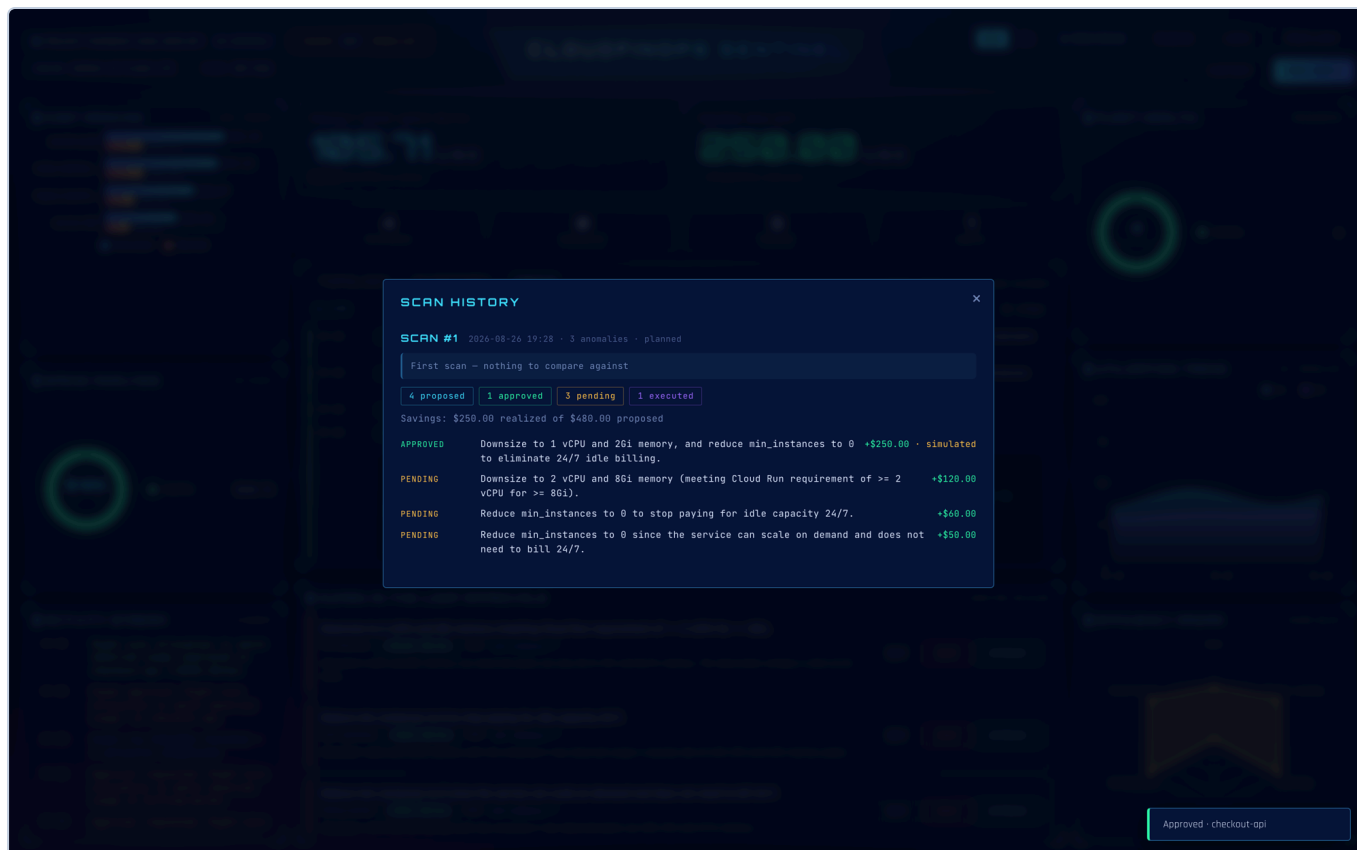
Execution payload

For a mutation the payload is the point. The trace records the approval, the handler that was dispatched, and the outcome. While `DRY_RUN=true` the step is recorded as `DRY_RUN - not sent to GCP` and carries the change it *would* have made, so the exact request can be reviewed before writes are ever enabled.

With `DRY_RUN=false` the same step carries GCP's own confirmation instead — the new revision name, the limits actually applied, and the `Ready` condition:

```
{
  "gcp_confirmed": true,
  "ready_state": "CONDITION_SUCCEEDED",
  "new_revision": "service-idle-00042-xyz",
  "applied_limits": {"cpu": "250m", "memory": "512Mi"},
  "uri": "https://service-idle-...run.app"
}
```

Scan history



Scan history

Each run is recorded with what it proposed, what was approved, what is still pending and what executed — and how much of the proposed saving was actually realized.

Trace messages, API method names and payload keys stay in English regardless of the UI language — it is a technical console, and translating `projects.locations.services.patch` would make it harder to match against GCP documentation.

What "real" means here

The product never presents invented infrastructure as real:

- **An empty project is a real answer.** A successful API call that returns zero services reports `data_source: gcp` with an empty fleet — it is not replaced with demo data.
- **Simulated data requires explicit opt-in.** Only `MOCK_MODE=true`, or an API failure with `ALLOW_SIMULATED_FALLBACK=true`, produces the demo fleet — and the header says `SOURCE: SIMULATED` when it does.
- **Unreadable sources are shown, not hidden.** A disabled API or missing role appears as an amber badge under the topology instead of silently reading as "nothing found".
- **Utilization is labelled.** `MONITORING` when it came from Cloud Monitoring, `MODELLED` when the metric was unavailable.
- **The two worlds never mix.** The on-ramp above is to run the simulated fleet first and point the agent at a real project afterwards. An approval ticket outlives the audit that raised it, so the demo fleet gets its own memory bank (`data/memory_bank.mock.json`), and every ticket records which world raised it. Approving a demo service against

a live project is refused rather than attempted — otherwise the agent tries to resize a service that never existed and the operator sees a bare `404 ... does not exist`.

Nothing runs until you ask

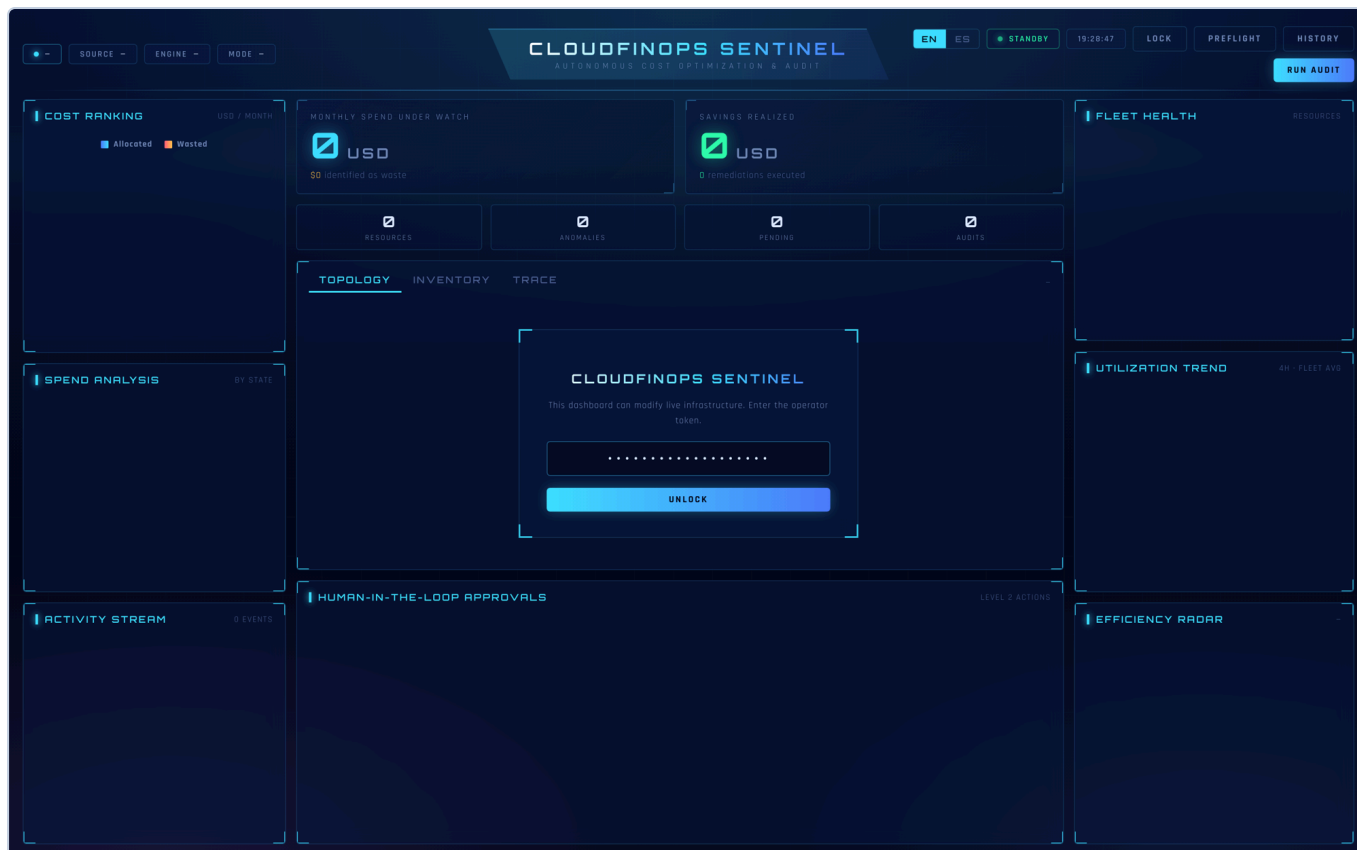


Idle state

Starting the service touches no GCP API. The dashboard opens idle (`SOURCE: NOT SCANNED`) and polling never triggers a scan — `/api/state` reads cache only. Discovery, analysis and remediation all begin when the operator presses **RUN AUDIT**, or a scheduler calls `/webhook/pubsub`.

That keeps a restart free, avoids surprise API quota usage, and makes the trace correspond to one deliberate run instead of ambient background activity.

Authentication



Login

There is no unauthenticated mode. Every endpoint that reads the estate or acts on it requires an operator token — locally and hosted alike. An "only in development" bypass is precisely what ends up deployed, and this dashboard deletes disks.

Start without a token and the agent generates one for that run and prints it; it never falls back to serving openly. On Cloud Run nothing is generated — the operator supplies it deliberately, and until then every request is refused.

The token is exchanged for an HttpOnly, SameSite=strict cookie; scripts can use `Authorization: Bearer <token>` instead. Comparison is constant-time. Cloud Scheduler authenticates with a **separate** `WEBHOOK_TOKEN`, so a leaked scheduler credential cannot press the approval buttons. `/health` stays open for Cloud Run probes and exposes nothing sensitive.

Spin-up instructions

Run it locally in two minutes

No API key and no GCP project required — it runs against a simulated six-service fleet and falls back to a deterministic heuristic audit if Gemini is not configured. The dashboard labels which mode you are in (`SOURCE: SIMULATED` , `ENGINE: HEURISTIC`) so demo data is never mistaken for live data.

```
git clone https://github.com/Mgodoyd/CloudFinOps-Sentinel-Agent.git
cd CloudFinOps-Sentinel-Agent

python3 -m venv .venv && source .venv/bin/activate
pip install -r requirements-dev.txt

cp .env.example .env          # add GEMINI_API_KEY to enable the LLM (optional)

MOCK_MODE=true DASHBOARD_TOKEN=dev-token \
  uvicorn app.main:app --reload --port 8080
```

Open <http://localhost:8080> and unlock it with `dev-token` . Press **RUN AUDIT**.

Run the tests with `pytest` — **305 pass, 2 skip** (the two need the OpenTelemetry exporter), no credentials needed.

Point it at a real GCP project

Drop your service-account JSON in the project root — `PROJECT_ID` and credentials are picked up automatically, no env var needed. Then check what actually works:

```
python -m app.tools.preflight
```

Preflight verifies credentials, every API the agent depends on, and write access, and prints the exact `gcloud` command that fixes each failure with your service account's email already filled in. The same report is at `/api/preflight` and behind the **PREFLIGHT** button in the dashboard.

Required APIs and roles

Capability	API	Role
Read Cloud Run services	<code>run.googleapis.com</code>	<code>roles/run.viewer</code>
Resize Cloud Run services	<code>run.googleapis.com</code>	<code>roles/run.admin</code>
Real CPU/memory utilization	<code>monitoring.googleapis.com</code>	<code>roles/monitoring.viewer</code>
Gemini via Vertex AI <i>(optional)</i>	<code>aiplatform.googleapis.com</code>	<code>roles/aiplatform.user</code>
Delete orphaned disks <i>(optional)</i>	<code>compute.googleapis.com</code>	<code>roles/compute.storageAdmin</code>
Purge untagged images <i>(optional)</i>	<code>artifactregistry.googleapis.com</code>	<code>roles/artifactregistry.admin</code>

```
PROJECT=your-project
SA=your-sa@your-project.iam.gserviceaccount.com

gcloud services enable run.googleapis.com monitoring.googleapis.com --project=$PROJECT
gcloud projects add-iam-policy-binding $PROJECT --member=serviceAccount:$SA --role=roles/run.viewer
gcloud projects add-iam-policy-binding $PROJECT --member=serviceAccount:$SA --
role=roles/monitoring.viewer
```

The DRY_RUN safety gate

`DRY_RUN=true` is the default. In that mode the agent does everything — detects anomalies, reasons, decides, opens approval tickets — but reports the change it *would* make instead of applying it. Remediations are recorded with `applied: false` and the header shows `MODE: DRY RUN`.

To let the agent actually modify infrastructure:

```
DRY_RUN=false uvicorn app.main:app --port 8080
```

The header switches to a red **LIVE WRITES** badge. In this mode `resize_cloud_run` performs a real read-modify-write on the service template (preserving image, env vars, scaling and concurrency) and deploys a new revision. Grant `roles/run.admin` first.

Deploying to Cloud Run

```
./deploy/deploy.sh
```

One command: enables the APIs, creates a least-privilege service account, provisions Firestore, stores the Gemini key and dashboard token in Secret Manager, builds and deploys the container, and schedules hourly audits with Cloud Scheduler. See [deploy/README.md](#) for what each role is for and how to verify the result.

The Cloud Scheduler job is what makes the agent autonomous rather than button-driven: it runs while nobody is watching and leaves approval tickets waiting.

Persistence

The Memory Bank is what stops the agent looping, so where it lives matters. `STATE_BACKEND=auto` picks the right one: **Firestore** when running on Cloud Run (whose filesystem is ephemeral — a new revision would otherwise forget everything it had already remediated), a **local JSON file** otherwise. Set it to `firestore`, `file` or `none` to override.

If Firestore is unreachable the agent starts anyway and says so: an agent that cannot read its history is still more useful than one that will not boot.

Configuration

Everything is environment-driven — see [.env.example](#). Credentials resolve through Application Default Credentials on Cloud Run.

Variable	Default	Purpose
PROJECT_ID / REGION	(from key file) / us-central1	Which project to audit
GEMINI_API_KEY	(empty)	Enables the LLM agent; empty → heuristic mode
USE_VERTEX	false	Use Vertex AI with the service account instead of a key
GEMINI_MODEL	gemini-3.5-flash-lite	Must support generateContent ; falls back automatically
DRY_RUN	true	Safety gate — false lets the agent mutate live infrastructure
USE_REAL_METRICS	true	Pull utilization from Cloud Monitoring
METRICS_LOOKBACK_HOURS	24	Observation window for utilization peaks
MIN_SAVINGS_THRESHOLD	5.0	Below this a finding is reported, never actioned
HIGH_RISK_ROI_THRESHOLD	40.0	Savings above which a human must approve
MEMORY_ANOMALY_THRESHOLD_GIB	1.0	Memory allocation that counts as oversized
MAX_TOOL_CALLS	4	Tool calls per audit; each one is an API request
METRICS_CACHE_TTL	60	Seconds a Cloud Run listing is cached
STATE_FILE	data/memory_bank.json	Local memory-bank path
STATE_BACKEND	auto	firestore on Cloud Run, file locally
DASHBOARD_TOKEN	(generated locally)	Required. Unlocks the dashboard and API
WEBHOOK_TOKEN	(falls back to dashboard)	Separate credential for Cloud Scheduler
OTEL_ENABLED	true	Export OpenTelemetry spans to Cloud Trace
MOCK_MODE	false	Skip GCP entirely and use simulated infrastructure
SCAN_REGIONS	auto	Regions to scan, or a comma-separated list
ALLOW_SIMULATED_FALLBACK	false	Substitute demo data when a GCP call fails

Multi-region and multi-service discovery

Scanning one region misses services deployed elsewhere. `SCAN_REGIONS=auto` probes ten common Cloud Run regions in parallel; set it to a comma-separated list to narrow the scan.

Discovery covers Cloud Run services, orphaned persistent disks, unused static IPs and untagged Artifact Registry images, all in one inventory. Each source degrades independently: a disabled Compute API does not stop the Cloud Run audit.

API

Method	Path	Description
GET	/	Command deck dashboard
GET	/health	Liveness probe + active agent mode
GET	/api/state	Full dashboard snapshot: KPIs, approvals, resources, charts, activity
GET	/api/resources?refresh=true	Fleet inventory, optionally bypassing the cache
GET	/api/resources/{id}/rationale	Why a resource was flagged, and the concrete fix
GET	/api/events?limit=50	Activity log
GET	/api/trace?since=N	Execution steps with request/response payloads
GET	/api/trace/stream	Live SSE feed of execution steps
POST	/api/trigger	Start an audit in the background
POST	/api/audit	Run an audit and wait for the result
POST	/api/approvals	{resource_id, status} — approve or reject a ticket
POST	/webhook/pubsub	Cloud Scheduler / Pub/Sub push endpoint
GET	/api/preflight	Readiness report: credentials, APIs, roles, write access
POST	/api/reset	Clear the memory bank (demo reset)

Interactive docs at `/docs`.

Live updates

The dashboard polls every 5 seconds, but waiting for the next tick makes a click feel unacknowledged. Every mutation — an approval, a rejection, a completed execution, a finished audit — pushes a `{"kind": "state"}` message down the same SSE stream that carries the trace, and every open dashboard refetches immediately. Measured approval-to-render latency: **~12 ms**, against a 5000 ms poll interval. The poll remains as a fallback, and the approval card dims optimistically so the click registers before the round-trip completes.

Resilience

The LLM is the optional part of this system. The audit itself — anomaly detection, cost math, the autonomy matrix, the memory bank — is deterministic and always runs. When Gemini is unavailable the agent finishes heuristically and says so, rather than losing the cycle:

Failure	Behaviour
No API key configured	Heuristic audit, <code>ENGINE: HEURISTIC</code> in the header
429 quota exhausted	Heuristic fallback + retry-delay hint
503 on every candidate model	Heuristic fallback, findings unaffected
Model not found (404)	Falls through <code>MODEL_FALLBACKS</code> automatically
Model at capacity (503)	Tries the next model — capacity is per-model
Cloud Run API unavailable	Error surfaced; simulated fleet only if explicitly enabled
Cloud Monitoring unavailable	Modelled utilization, labelled <code>MODELLED</code> , 5-min backoff

Degraded runs are marked in the audit report with an amber banner explaining what was skipped.

`audit_infrastructure()` always returns the same keys, on success and on failure alike.

Model availability

Google retires models for *new* API keys while keeping them alive for existing ones, so a model can appear in `models.list()` and still return **404 NOT_FOUND** when you call it — `gemini-2.5-flash` behaves exactly this way for recently created keys.

A 404 here means "this model is not available to your key", not "your key is invalid". Preflight distinguishes the two and, on a 404, lists the models your key can actually call instead of sending you to regenerate a working credential.

Choosing a model

Model family	Supported action	Usable here
<code>gemini-*-flash</code> , <code>gemini-*-flash-lite</code>	<code>generateContent</code>	✓
<code>gemini-*-live-*</code> , <code>*-translate-*</code>	<code>bidGenerateContent</code>	✗ Live API over WebSocket
<code>*-image</code> , <code>*-tts</code> , <code>veo-*</code> , <code>lyria-*</code>	media generation	✗

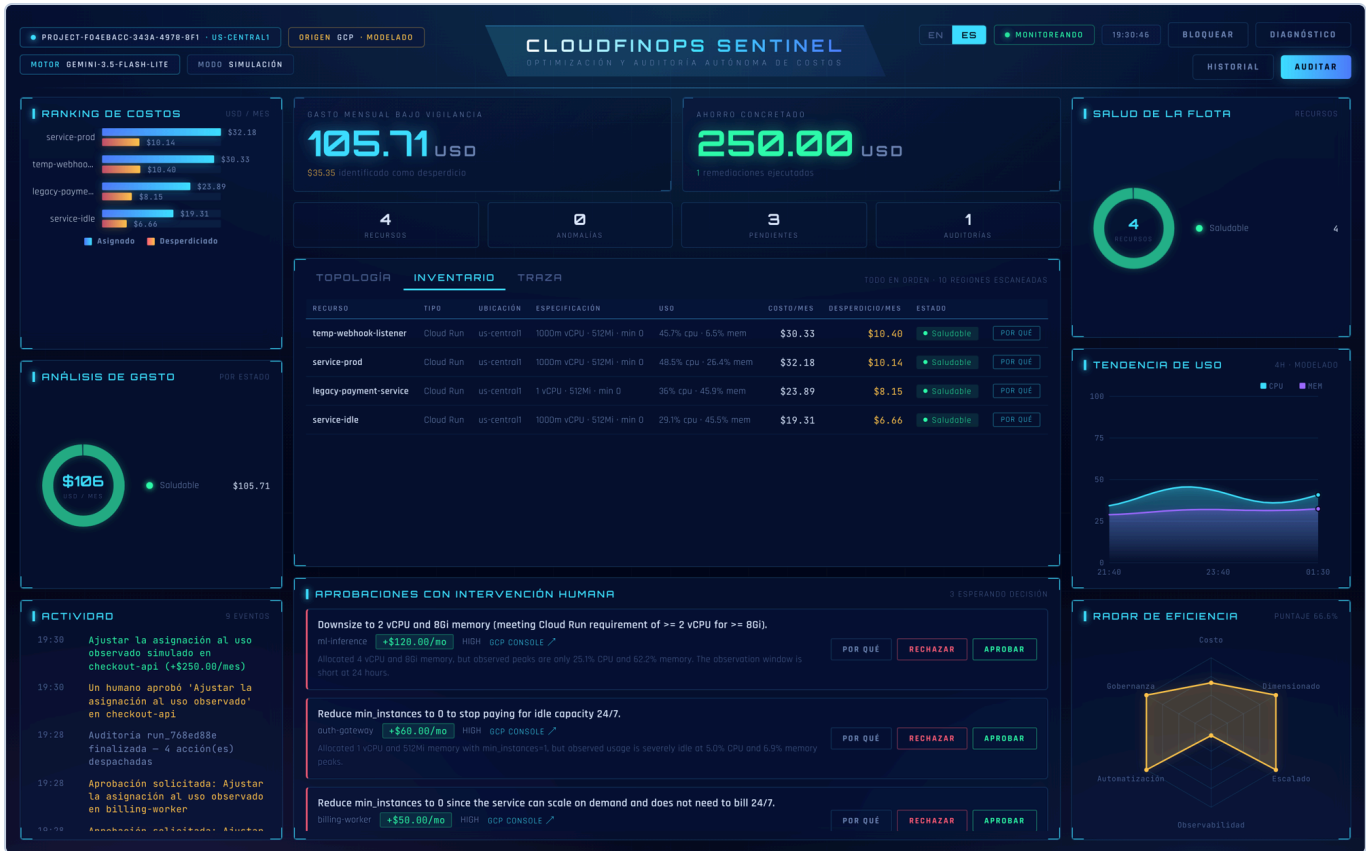
A Live model is not a lighter version of Flash — it is a different protocol for streaming audio and video. Configuring one produces a failure that looks like a broken credential, so preflight checks `supported_actions` first and names the real reason.

`flash-lite` is the default: it has the highest free-tier request limit and returns the fleet analysis in about two seconds, where the heavier Flash models frequently time out on a fleet-sized prompt.

Gemini free-tier quota

The free tier allows **5 requests per minute**. `MAX_TOOL_CALLS` defaults to `4` so a single audit fits inside that budget. On a paid tier, raise it with `MAX_TOOL_CALLS=12`.

Languages



Inventory in Spanish

The interface ships in English and Spanish, switchable from the header. The choice persists in `localStorage` ; on first visit it follows the browser's `Accept-Language` , falling back to English.

Translation spans three layers, because UI text comes from three places:

Layer	Where it lives	How it is translated
Static chrome	<code>static/js/i18n.js</code>	<code>data-i18n</code> attributes, swapped in place
Generated analysis	<code>core/i18n.py</code>	<code>/api/state?lang=</code> returns evidence, rules, solutions and autonomy already localised
Persisted text	Memory bank	Events and approval tickets store a key plus structured params , rendered at read time

That third layer matters: an approval ticket raised during a nightly audit is read later, possibly by someone using the other language. Storing the finished sentence would freeze it in whichever language the agent happened to run in, so tickets persist `action_key` and `change_specs` (`{kind: "memory", from: "2Gi", to: "512Mi"}`) instead of prose.

Figures never change across languages — a test asserts every number in the evidence table is identical in both. `gcloud` commands are never translated. Resource `status` stays a stable English token for CSS and filtering; only the human-facing `verdict` is localised.

Tests

```
pytest
```

307 tests — 305 pass and 2 skip where the OpenTelemetry exporter is unavailable — covering the memory bank, cost math, the autonomy matrix, the DRY_RUN safety gate, tool serialization, LLM analysis and failure handling, model fallbacks, action dispatch per resource type, the real-data guarantees, the rationale engine, translation completeness, the execution trace, scan history, lazy startup, preflight, the approval-to-execution contract, simulated / real isolation and the API — all against simulated infrastructure with writes disabled, no credentials required.

Project layout

app/	
main.py	FastAPI routes; blocking GCP calls run off the event loop
core/	
config.py	Environment-driven settings
i18n.py	Translation catalogue for generated text
trace.py	Execution trace + live SSE fan-out
analyst.py	One structured LLM call: judgement over the whole fleet
planner.py	Turns the analysis into an ordered plan
executor.py	Carries the plan out, enforcing the autonomy matrix
agent.py	Gemini client, tool loop, heuristic fallback
prompts.py	System instruction + audit prompt template
telemetry.py	OpenTelemetry setup → Cloud Trace
auth.py	Token auth, constant-time comparison
models/schemas.py	Pydantic contracts
tools/	
gcp_metrics.py	Cloud Run inventory, cost math, chart aggregations
gcp_inventory.py	Multi-region, multi-service real resource discovery
gcp_monitoring.py	Real CPU/memory utilization from Cloud Monitoring
gcp_actions.py	Real mutations, gated by the DRY_RUN safety flag
gcp_remediator.py	Agent-facing tools + autonomy matrix enforcement
rationale.py	Evidence, rules, sizing and the autonomy explanation
preflight.py	Readiness diagnostics ("what do I still need?")
memory_tools.py	Persistent memory bank (approvals, history, events)
state_store.py	Firestore / file / in-memory persistence backends
web/	
index.html	Command deck
static/css/js	Hand-built HUD; no framework, no CDN JS
static/js/i18n.js	UI string catalogue (en/es)
deploy/	One-command Cloud Run deploy + Cloud Scheduler
docs/	Architecture notes, diagrams and screenshots
tests/	307 tests, no credentials needed

Known limitations

- Cost is estimated from *allocated* CPU/memory at Cloud Run Tier-1 on-demand rates, assuming an always-on instance. It is not read from the Cloud Billing export, so treat it as a right-sizing signal rather than an invoice.
- Disk deletion and Artifact Registry purging are implemented but need their own APIs enabled and their discovery paths wired to your registry layout.

- When Cloud Monitoring is unreachable the dashboard falls back to a deterministic utilization model and labels itself `MODELLED` — never presenting simulated numbers as real ones. Confidence drops to `low` accordingly.

License

MIT — see [LICENSE](#).